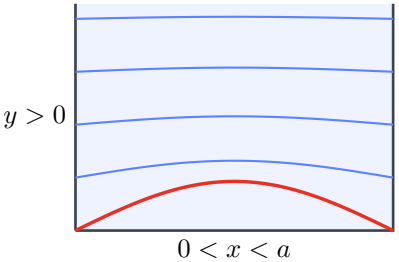
半无限条带的单模边界：横向正弦模态沿纵向指数衰减



曲线为不同高度的电势剖面示意

$$X'' + k^2 X = 0, \quad X(0) = X(a) = 0 \Rightarrow k_n = n\pi/a$$

边界只有 $n = 1$ 模态，因此无需叠加其余本征函数。

$$\varphi(x,y) = V_0 \sin\frac{\pi x}{a} e^{-\pi y/a}$$